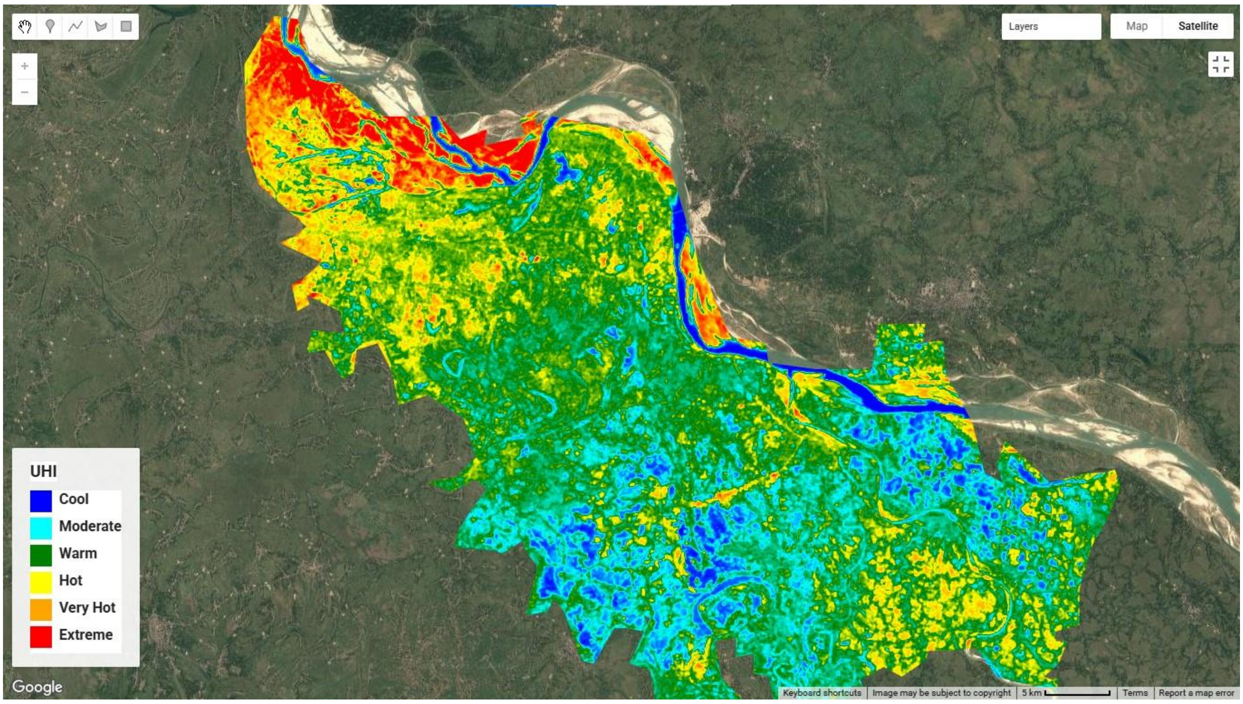
Exploring Urban Heat Island (UHI) Dynamics with Google Earth Engine




Over the past few days, I worked on a geospatial analysis project to investigate Urban Heat Island intensity in Kushtia, Bangladesh using Landsat 8 thermal imagery and Google Earth Engine.

This project helped me better understand how satellite-based thermal remote sensing can be used to monitor urban environmental conditions and surface temperature variations.

Project Highlights:
• Landsat 8 image acquisition and preprocessing
• NDVI-based vegetation analysis
• Surface emissivity estimation
• Brightness Temperature (BT) extraction
• Land Surface Temperature (LST) calculation
• Urban Heat Island (UHI) intensity mapping
• GeoTIFF export workflows
• Custom visualization in GEE

- Study Area: Kushtia, Bangladesh
- Time Period: March–April 2024

Through this workflow, this analysis observed:

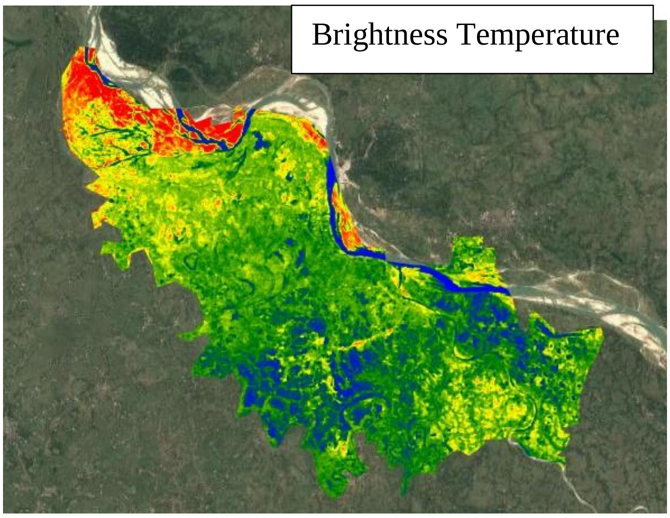
Higher temperatures concentrated in built-up and exposed land areas, cooler surface temperatures in vegetated zones and river and waterbody regions acting as thermal sinks.

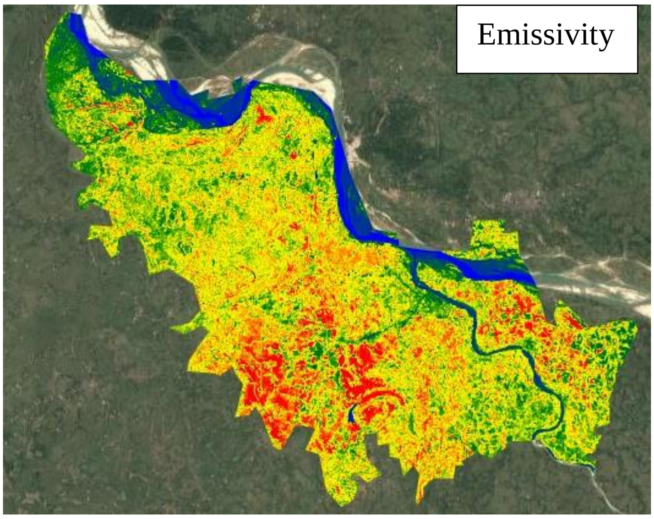
The project demonstrates how cloud-based geospatial platforms like Google Earth Engine can simplify large-scale satellite image processing and environmental monitoring. Working with thermal bands, emissivity correction, and surface temperature retrieval provided practical insights into real-world remote sensing applications.

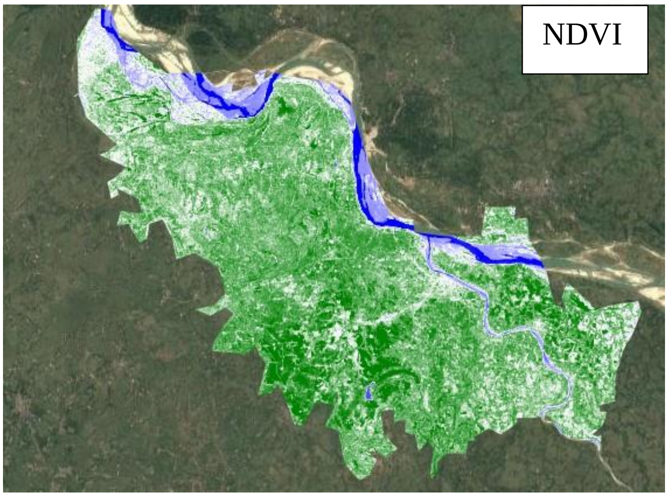
Technologies & Skills Applied:
• Google Earth Engine (GEE)
• Remote Sensing
• GIS & Spatial Analysis
• Landsat 8 Thermal Band Processing
• Spatial Data Visualization

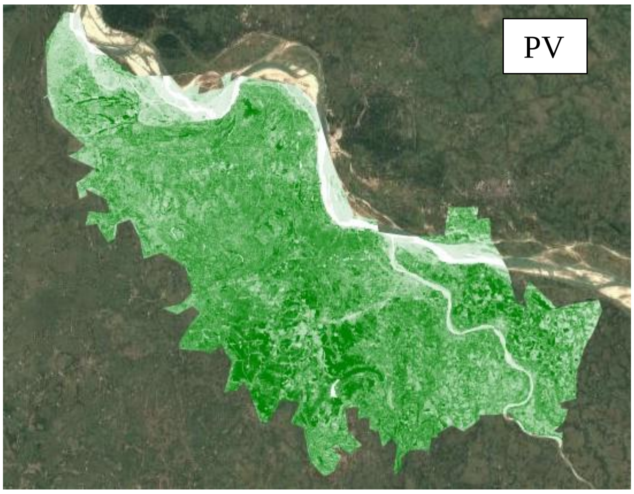
One of the most interesting aspects of this project was implementing the full LST workflow directly within Earth Engine — from preprocessing satellite imagery to generating Urban Heat Island maps and exporting analysis-ready datasets.

This project also strengthened my understanding of:
✓ Thermal remote sensing concepts
✓ Land surface temperature retrieval methods
✓ Spatial pattern analysis
✓ Geospatial data processing in cloud environments




NDVI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